ALUMINIUM HYDROXIDE + MAGNESIUM HYDROXIDE

Separate from other drugs by 2 h

Balanced bowel effect

CHEMICAL NAME $Al(OH)_3$ + $Mg(OH)_2$ (often with simeticone)
CLASS Antacid combination
TRADE NAMES Maalox · Mylanta (with simeticone) · Gelusil · Mucogel

THERAPEUTIC USES / INDICATIONS

- Heartburn, dyspepsia, acid indigestion.
- Symptomatic relief of GORD and peptic ulcer pain.
- Flatulence (with simeticone).

PHARMACODYNAMICS

- Neutralise gastric **hydrochloric acid** → ↑ gastric pH; ↓ pepsin activity.
- Mg → laxative; Al → constipating; combination balances bowel effects.

ADMINISTRATION

- **PO** suspension (shake well) or chewable tablets 1–3 h after meals and at bedtime, or when symptoms occur.
- Separate from other oral drugs by 2–4 h (binds them).

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Diarrhoea (Mg) or constipation (Al).
- Chalky taste.
- Belching.

ADVERSE EFFECTS (SERIOUS)

- Hypermagnesaemia, aluminium toxicity (encephalopathy, osteomalacia) in renal failure.
- Hypophosphataemia (Al binds phosphate).
- Milk–alkali syndrome (with calcium).

DOSAGES

- **Adult:** 10–20 mL suspension or 1–2 tablets 4 times daily (after meals and at bedtime).
- **Children ≥12 yr:** adult dose; younger on medical advice.
- Short-term use unless supervised.

CONTRAINDICATIONS & CAUTIONS

- Severe renal impairment (relative).
- Hypophosphataemia.
- Undiagnosed GI bleeding; appendicitis.

KEY INTERACTIONS

- Reduce absorption of tetracyclines, fluoroquinolones, iron, digoxin, levothyroxine, bisphosphonates, azoles, enteric-coated tablets (premature release).
- **Citrate:** ↑ Al absorption.

PHARMACOKINETICS

- **Onset:** minutes. **Duration:** 30 min (empty stomach) to 3 h (after meals).
- Minimal absorption of Al and Mg (accumulate in renal failure).
- Excreted in faeces.

NURSING MANAGEMENT

- Schedule doses away from other medications.
- Monitor bowel pattern.
- Monitor Mg^{2+} and phosphate in renal impairment.
- Persistent symptoms: refer for investigation.

PATIENT EDUCATION

- Shake well; take 1 h after meals and at bedtime.
- Keep 2–4 h apart from other medicines.
- See a doctor if symptoms last >2 weeks or you have weight loss/vomiting.

KEY POINTS

- Antacid; neutralises acid.
- Binds many drugs — separate by 2–4 h.
- Mg = diarrhoea, Al = constipation.
- Caution in renal failure.